

Title: Train Coach Management Using Circular Linked List in C++.

Problem Statement:

Design and Implement a Train Coach Management System in C++ using a circular linked list. Each node represents a train coach and stores its coach number. The system allows users to add new coaches, removes existing coaches, and display all coaches connected to the train. The circular linked list maintains the connecting between the last coach and the first coach, representing the circular arrangement of coaches.

Objectives:

- To implement a train coach management system using C++.
- To understand the concept of a circular linked list.
- To add new coaches to the train.
- To remove the coaches from the train.
- To display all available coaches in sequence.
- To understand dynamic memory

- and pointer manipulation.
- To perform insertion and deletion operations efficiently.

Algorithm:

Step 1: Start the program.

Step 2: Initialize an empty circular linked list by setting last=NULL. →

Step 3: Display the menu containing the following operations.

1. Add a coach.
2. Remove a coach.
3. Display all coaches.
4. Exit.

Step 4: Accept the user choice. →

Step 5: If the user selects Add coach:

- Create a new node.
- Enter the coach number.
- If the list is Empty, point the new node to itself.
- Otherwise, insert the new coach at the end update the circular connection.

Step 6: If the user selects Remove Coach:

- Enter the coach number to remove.
- Search for the coach in the list.
- If found, delete the node and update the links.
- If not found, display an appropriate message.

Step 7: If the user selects Display Coaches, traverse the circular linked list & display all coach numbers.

Step 8: Repeat steps 3-7 until the user selects Exit.

Step 9: Release the allocated memory and stop the program.

flow Chart.

Start

Initialize circular linked list (last = NULL).

Display Menu.

1. Add Coach.
2. Remove coach
3. Display coaches
4. Exit.

Read Choice

Choice ?

Read coach number

Is list empty?

Yes No

Read coach number to remove

Is list empty.

Yes No

Traverse the list from last $\rightarrow$ next.

Display all coach numbers in sequence
Also show "back to first coach".

Exit program

Stop

